Mining Massive Data Sets
Problem Set: Frequent Itemsets, LSH, and Random Projection

Instructions

For modeling problems, justify the representation and similarity/mining choices you make. Unless a problem explicitly specifies an algorithm, choose an appropriate method from the course and explain why it fits the setting.

1 Frequent Itemset Mining

Problem 3: Web Browsing

A website records every page visit as

$$(\text{user ID}, \text{URL}, \text{timestamp}).$$

The website does not explicitly record browsing sessions. The company wants to identify groups of pages that users commonly visit during the same browsing session. Assume that two consecutive page visits from the same user belong to different sessions if they are separated by more than 30 minutes.

Describe how frequent-itemset mining can be used to discover these patterns.

Problem 6: Software Development

A software company maintains the complete Git history of a large project. For every commit, the system records the developer, timestamp, and files that were added, deleted, or modified. The development team wants to discover groups of files that are frequently changed together. Such patterns could help identify dependencies between components and assist developers when reviewing code changes.

Describe how frequent-itemset mining can be applied.

Problem 9: Online Shopping Behavior

An online retailer stores

$$(\text{customer ID}, \text{product ID}, \text{order ID}, \text{timestamp}).$$

The retailer is interested in a longer-term pattern than ordinary market-basket analysis. It wants to identify groups of products that are commonly purchased by the same customer within the same calendar month, even if those products belong to different orders.

Describe how frequent-itemset mining could be used to discover such patterns.

Problem 12: Online Learning

A large online learning platform records student activities such as watching a lecture, reading course material, attempting an exercise, participating in a discussion, and completing a quiz. Every activity has a student ID, course ID, activity type, resource ID, and timestamp.

The platform wants to discover groups of learning resources that successful students commonly use during the seven days before completing an assessment. The goal is to identify useful combinations of learning materials that could later be recommended to other students.

Propose a frequent-itemset mining formulation for this problem and explain how the discovered patterns could be interpreted.

2 Locality-Sensitive Hashing and Random Projection

Problem 13: Near-Duplicate Web Pages

A search engine has crawled several billion web pages. Many pages are almost duplicates because of mirrors, copied articles, templates, advertisements, navigation menus, or small edits. The search engine wants to identify pairs of pages whose textual contents are highly similar. Comparing every pair of pages is computationally infeasible.

Design a scalable method for finding candidate pairs of near-duplicate pages. Explain how the pages should be represented, how similarity should be measured, and how the number of pairwise comparisons can be greatly reduced.

Problem 17: Choosing LSH Parameters

A system contains 10 million documents. After preprocessing, each document is represented by a MinHash signature containing 120 values. The system uses an LSH scheme in which the signature is divided into b bands containing r rows each, where

$$br = 120.$$

Consider

$$(b, r) = (40, 3), (30, 4), (24, 5), (20, 6), (15, 8).$$

The system should retrieve document pairs whose Jaccard similarity is approximately 0.8 or higher, while avoiding as many low-similarity candidate pairs as possible.

Select an appropriate configuration and justify your choice quantitatively. Compare the probability that pairs with similarities

$$0.3, \quad 0.5, \quad 0.7, \quad 0.8, \quad 0.9$$

become candidate pairs.

Problem 18: False Positives and False Negatives

An LSH-based plagiarism-detection system is designed to identify documents with Jaccard similarity of at least 0.75. After deployment, instructors report two problems: too many unrelated document pairs are being sent for expensive comparison, and occasionally genuinely similar documents are missed entirely. The system uses MinHash signatures followed by banding.

Recommend changes to the system that could address each problem. Discuss the trade-off between reducing false positives and reducing false negatives.

Problem 19: High-Dimensional Document Vectors

A document-search system represents each document using a 100,000-dimensional vector derived from its textual content. The collection contains 20 million documents. Computing and storing similarities directly in the original space is expensive. The company would like to replace each vector with a much smaller vector while approximately preserving distances between documents.

Design an approach for reducing the dimensionality to 1,000 dimensions. Explain why the resulting vectors can still be useful for similarity search and what information may be lost.

Problem 21: How Many Dimensions?

A dataset contains n high-dimensional points. The system wants to map these points into a lower-dimensional Euclidean space while approximately preserving pairwise distances. Consider

$$n = 10^3, \qquad n = 10^6, \qquad n = 10^9.$$

The system can tolerate a multiplicative distortion of approximately $1 \pm \epsilon$.

Using the theoretical relationship between the required projected dimension, n, and ϵ, discuss how the required dimension changes when (i) the number of data points increases and (ii) the acceptable distortion decreases. Explain why this result is useful for massive datasets whose original dimensionality may be extremely large.

Problem 22: Constructing a Random Projection

Four objects are represented by vectors

$$x_1, x_2, x_3, x_4 \in \mathbb{R}^{1000}.$$

They must be transformed into

$$y_1, y_2, y_3, y_4 \in \mathbb{R}^{50}.$$

A random matrix

$$R \in \mathbb{R}^{50\times 1000}$$

is available.

Describe how the transformation should be performed, including any appropriate normalization of R. After projection, explain how you would experimentally determine whether the transformation has preserved the geometry of the original four objects sufficiently well.

Problem 23: Image Similarity Search

An image-sharing service stores 500 million images. A neural network converts each image into a 2,048-dimensional real-valued embedding. Images depicting similar visual content tend to have vectors pointing in similar directions. Given a query image, the service needs to quickly retrieve potentially similar images without comparing the query against all 500 million vectors.

Design a scalable approximate similarity-search system. Explain how high-dimensional vectors can be mapped into compact signatures or buckets such that similar images have a high probability of being considered together. Explain why the approach is approximate rather than exact.

Problem 24: Random Hyperplanes

Suppose two normalized vectors x and y form an angle θ. A random vector r is generated, and each vector is represented by one bit:

$$h_r(x) = \begin{cases} 1, & r \cdot x \geq 0, \\ 0, & r \cdot x < 0. \end{cases}$$

Derive the probability that $h_r(x) = h_r(y)$ as a function of θ. Explain what happens when the vectors are almost parallel, orthogonal, or point in opposite directions.

The system then generates 100 independent random vectors and produces a 100-bit signature for each object. Explain how these signatures could be used to efficiently identify potentially similar objects in a collection containing millions of objects.